

Reviewer Pack — Minimal Rank of Twisted Hodge Classes vs DPLL Proof Length

Entry 418d073e7755 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Twisted Hodge Classes vs DPLL Proof Length

Entry ID: 418d073e7755 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 00:17:03 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Twisted Hodge Theory) **Field B** (complexity object): Complexity Theory: DPLL Proof Complexity

Statement:

{‘quantitative_relation’: ‘ $E[\text{Rank}(\text{TwistedH}(n))] = \Theta(f(n))$ for all instances $n \leq 40$, where $\text{Rank}(\text{TwistedH}(n))$ is the minimal rank of a twisted Hodge class over the polynomial algebra of n variables and $f(n)$ is an increasing function.’, ‘falsifier’: ‘There exists an instance $n \leq 40$ such that $\text{Rank}(\text{TwistedH}(n)) > \Theta(f(n))$.’, ‘constructive_mapping’: ‘For an instance of size n , construct the twisted Hodge class by computing the cup product of a generic line bundle with itself k times, where k is chosen such that the resulting class has minimal rank among all possible choices.’}

Rationale (proposer’s reasoning):

{‘exposure_of_structure’: ‘Twisted Hodge theory provides a rich algebraic structure that could potentially reveal hidden complexities in proof length, offering new insights into the structure of DPLL proofs.’, ‘link_to_complexity’: ‘The minimal rank of a twisted Hodge class is expected to correlate with the depth of the search tree in a DPLL proof, as it reflects the complexity of the underlying algebraic structure.’}

Taxonomy category: Geometric_Invariant_vs_Proof_Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 871033457582cd24

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For $n \leq 40$, if the minimal rank of Twisted Hodge classes ($\text{Rank}(\text{TwistedH}(n))$) is within $\Theta(f(n))$ for at least 80% of the instances and the mean $\text{Rank}(\text{TwistedH}(n))$ is less than or equal to 3 across all seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'Twisted Hodge Theory' AND 'DPLL proof length' AND 'quantitative relation' - 'algebraic geometry' AND 'proof complexity' AND 'twisted Hodge classes' AND 'minimal rank' - 'polynomial algebra' AND 'cup product' AND 'generic line bundle' AND 'DPLL proof complexity'

Top relevant hits considered: - [<http://arxiv.org/abs/1101.3647v1>] Notes on absolute Hodge classes - [<http://arxiv.org/abs/math/0312156v1>] Some Hodge theory from Lie algebras - [<http://arxiv.org/abs/1601.00001>] Pink's theory of Hodge structures and the Hodge conjecture over function fields - [<http://arxiv.org/abs/0912.2255v1>] Test ideals via algebras of p^{-e} -linear maps - [<http://arxiv.org/abs/math/0502387v3>] Multiplier ideals in algebraic geometry - [<http://arxiv.org/abs/1801.10489v3>] Hodge level for weighted complete intersections - [<http://arxiv.org/abs/1701.07768v2>] Cup products, lower central series, and holonomy Lie algebras - [<http://arxiv.org/abs/math-ph/0603053v1>] Clifford algebra-parametrized octonions and generalizations

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=9.4s

5.1 Generated Python source

```
import random
from fractions import Fraction
import math

def gaussian_elimination(A):
    rows, cols = len(A), len(A[0])
    for i in range(rows):
        max_row = i + max(range(i, rows), key=lambda j: abs(A[j][i]))
        A[i], A[max_row] = A[max_row], A[i]
        if A[i][i] == 0:
            return None
        for j in range(cols):
            A[i][j] /= A[i][i]
        for k in range(rows):
            if k != i and A[k][i] != 0:
                factor = A[k][i]
                for j in range(cols):
                    A[k][j] -= factor * A[i][j]
    return [sum(row) for row in A]
```

```

def twisted_hodge_class(n, k):
    H = [[Fraction(1 if i == j else 0, 1) for j in range(n)] for i in range(n)]
    for _ in range(k):
        new_H = []
        for i in range(n):
            new_row = [sum(H[i][j] * H[j][k] for k in range(n)) for j in range(n)]
            new_H.append(new_row)
        H = new_H
    rank = gaussian_elimination(H)
    if rank is None:
        return 0
    return sum(1 for r in rank if r != 0)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        rank = twisted_hodge_class(n, random.randint(1, 5))
        if rank is None:
            return {
                "metric_name": "Rank(TwistedH(n))",
                "metric_value": float('inf'),
                "instances_tested": len(n_values),
                "conjecture_holds": False,
                "counterexample": "gaussian_elimination failed"
            }
        results.append(rank)

    mean_rank = sum(results) / len(results)
    conjecture_holds = all(r <= 3 for r in results)
    counterexample = "" if conjecture_holds else f"Rank(TwistedH(n)) > 3 for some n"

    return {
        "metric_name": "Rank(TwistedH(n))",
        "metric_value": mean_rank,
        "instances_tested": len(results),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else list(range(2, 80, 3))

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result["metric_value"])

    mean_rank = sum(results) / len(results)
    support_fraction = sum(r <= 3 for r in results) / len(results)

    if all(r <= 3 for r in results):

```

```

    print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(i for i, r in enumerate(results) if r > 3)
    print(f"RESULT: FALSIFIED counterexample=\"Rank(TwistedH(n)) > 3\" first_failing_seed={seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

ure_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Rank(TwistedH(n))', 'metric_value': 0.0, 'instances_tested': 6, 'conjecture_
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TRIAL: {'metric_name': 'Rank(TwistedH(n))', 'metric_value': 0.0, 'instances_tested': 6, 'conjecture_
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test has been conducted on only 6 instances, which is too small to draw a general conclusion about the conjecture. The metric does not necessarily scale trivially with n , and there is no indication that the tested range of n covers the full scope of the conjecture's claim.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test has been conducted on only 6 instances, which is too small to draw a general conclusion about the conjecture. The critic challenged the validity of the results due to the limited number of tested instances and the potential for the metric not scaling trivially with n . | next: Increase the number of instances tested to at least 80% of the total ($n \leq 40$) and ensure that the mean Rank(TwistedH(n)) is less than or equal to 3 across all seeds before drawing a conclusion.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11983
2	preregistration	ollama_remote	glm4:latest	0	0	5982
3	novelty	ollama_remote	glm4:latest	0	0	4937
4	novelty	ollama_remote	glm4:latest	0	0	6335
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	29175
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7759
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10462
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9696
9	critic	ollama_remote	glm4:latest	0	0	10028
10	judge	ollama_remote	glm4:latest	0	0	8004

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 104361 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/418d073e7755.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/418d073e7755.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/418d073e7755.tar.gz* (if generated)